non-Hausdorfness

This is a short blog to explain and construct a non-Hausdorff spaces to someone with no university background in mathematics. As a one-sentence recap:

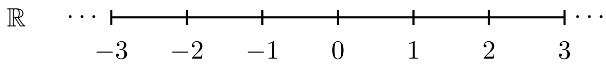
A non-Hausdorff space is a space where points can be infinitely close to each other without being the same point

There shall be some small apocrypha's for the sake of not overloading on technicalities, and these shall be sorted in an extra technicalities section.

This is an object that is usually introduced in a class in topology and relies on a lot of little ideas. These ideas are presented here with as little build-up as possible, but as a consequence it is likely will have to spend some time digesting the concepts.

1 Build up

The set of all numbers that are not complex are called the *real number line* and is denoted $\mathbb{R}$. It is often drawn as:



An element of $\mathbb{R}$ is called a *real number* (to distinguish from complex numbers). Let us choose any two real numbers, say 3 and 4. Then they are a distance 1 apart from each other. Any two *distinct* real numbers x, y have some distance, namely $x - y$ or $y - x$ (depending on which number is bigger). If the distance distance between two real numbers is 0, then they are the same number, since:

$$x - y = 0 \qquad \text{then} \qquad x = y$$

But there are spaces in which you can have two points that have *zero distance* between them and yet are *distinct*! It was a mathematician named *Hausdorff* that first stumbled upon such a weird space, and thus in his honour a space in which there can't exist distinct points that have zero distance are called *Hausdorff spaces*, and the weird spaces are called *non-Hausdorff* otherwise. For example the real numbers $\mathbb{R}$ is Hausdorff.

2 What is a Space?

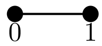
What is meant here by "space"? The first encounter we all have with the notion of "space" is within the world we live in: we consider the "space" of a room, or the routes of a hiking trail the "space" on which we can walk. These two examples of spaces are very different from each other (in a room, we have 3 directions, while for hiking trails we usually are limited to lines to follow), and there are many more such examples that come up in the sciences (for example, we consider all possible outputs of a experiment the "total space" of outcomes). This generality of the notion of space has lead mathematics to define the notion of *topology* which is a vast generalization of the Cartesian plane. Going through the technicalities of defining a topology would distract from the specific goal of constructing a non-Hausdorff space. In fact, going through the technicalities of defining spaces in higher than 1 dimension would increase the technicalities, and we can construct a 1-dimensional space that is not Hausdorff. Thus, for our purposes, we shall define a space as:

A set containing numbers and possibly containing sets of numbers

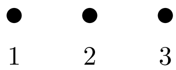
This is very abstract, so let us immediately justify this definition. The simplest space is $\mathbb{R}$ which is just the set of all numbers and represents a line. We can take line segments from it to recover finite lines, for example:

$$[0, 1] = \{x \text{ in } \mathbb{R} \text{ such that } 0 \leq x \leq 1\}$$

which we can visualize as:



This is an infinite set as it contains all numbers between 0 and 1 (including 0 and 1). We can also take a finite subset, for example the definition of a 1-dimensional space allows for subset of numbers of $\mathbb{R}$, for example $\{1, 2, 3\}$ is three numbers and can be represented as three points:



This covers the first case of the definition "a set containing numbers". Let us now justify the 2nd part of the definition: "possibly containing sets of numbers". To understand this better, let us start by being precise what we mean by *set*. A general set is simply the way mathematicians group things, for example here are two different sets, one containing the numbers $1, 2, 3$ and the other the numbers $3, 4, 5$:

$$\{1, 2, 3\} \qquad \text{and} \qquad \{3, 4, 5\}$$

where we used the $\{,\}$ notation to enclose the elements we are considering. Sets can have more than numbers, for an arbitrary example a set can contain all the words in the English dictionary:

$$S = \{\text{..., Hamiltonian, characteristic, Hypothesis, ...}\}$$

A set can also contain other sets (think by analogy a warehouse that has boxes containing stuff):

$$W = \{\{\text{Alice in Wonderland, Introduction to Analysis}\}, \{\text{spoon, fork, knife}\}\}$$

For our purposes, we have sets containing numbers and sets of numbers, for example:

$$R = \{1, 2, \{3, 4\}\}$$

This is exactly what we are calling a space. Next it must be justified why numbers and sets of numbers are being mixed. The idea is that the numbers should be considered as points, and the sets of numbers should be thought of as being points that are *glued together*.

Principle of Gluing

Let us demonstrate this with an example: the following set splits the even and the odd numbers into two sets:

$$P = \{\{0, 2, 4, 6, ...\}, \{1, 3, 5, 7, ...\}\} = \{E, O\}$$

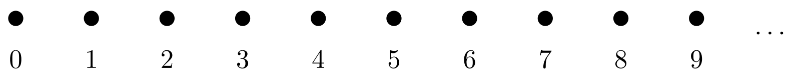
where we labeled the sets within P as E for even numbers and O for odd numbers. Notice that P only has two sets (naturally each set contains infinitely elements). What we shall do now is show how we can interpret the fact that P has 2 sets as all the elements within E and O respectively "equal" to each other, that is:

$$0 = 2 = 4 = \cdots \qquad \text{and} \qquad 1 = 3 = 5 = \cdots$$

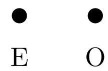
Here is how to think of this equality: the set P *only* remembers the *parity* of each number. It does not care if a number is bigger or smaller or divisible by 5, it only cares if it is even or odd. The equality in the above equation is an *equality of parity*, meaning two numbers are the same if they have the same parity. Using the equal sign "=" is confusing as we already use it to mean to numbers are "exactly equal" not just "equal in parity" (or as a mathematician would say, equal *up to parity*). Let us thus create a new symbol $\sim_P$ and write:

$$0 \ \sim_P \ 2 \ \sim_P \ 4 \ \sim_P \cdots \qquad \text{and} \qquad 1 \ \sim_P \ 3 \ \sim_P \ 5 \ \sim_P \cdots$$

So $a \ \sim_P \ b$ is both a and b are even, or both a and b are odd, and otherwise we write $a \ \not\sim_P \ b$. We can go one step further and say that we "glued" these numbers together. If we represent the natural numbers $\{0, 1, 2, ..\}$ as points we get:



Then the set P that glued the evens and odd can visually be represented as:



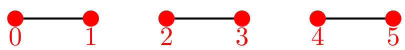
Then the set P glued all the even points together and all the odd points together, resulting in two points. In other words, we should treat $\{0, 2, 4, ...\}$ and $\{1, 3, 5, ...\}$ as single *points*!

Using gluing to form a triangle

Thinking of the above process as gluing is actually how mathematicians construct many objects. To show this is not just a trick, let us show how we can create a triangle using this method. Say we had three straight lines L_1, L_2, L_3 each of length 1. Let us represent each as intervals:

$$\begin{aligned} L_1 &= [0, 1] = \{x \text{ in } \mathbb{R} \text{ such that } 0 \leq x \leq 1\} \\ L_2 &= [2, 3] = \{x \text{ in } \mathbb{R} \text{ such that } 2 \leq x \leq 3\} \\ L_3 &= [4, 5] = \{x \text{ in } \mathbb{R} \text{ such that } 4 \leq x \leq 5\} \end{aligned}$$

Notice that we are taking x in the real numbers; for example in L_1 all numbers between 0 and 1 (including 0 and 1) are in L_1 (ex. 0.5 and $\pi - 3$ are in L_1). Visually, these three sets represent lines of length 1:



Let us now glue these lines to form a triangle, we want to:

1. glue 1 in L_1 to 2 in L_2: this is represented by the set $\{1, 2\}$ which is treated as a single point
2. glue 3 in L_2 to 4 in L_3: this is represented by the set $\{3, 4\}$ which is treated as a single point
3. glue 5 in L_3 to 0 in L_1: this is represented by the set $\{5, 0\}$ which is treated as a single point

The rest of the points of the lines L_1, L_2, L_3 will be left alone. Let us separate the end-points of the lines so that we may glue them. Let:

$$E_1 = (0,1) = \{x \text{ in } \mathbb{R} \text{ such that } 0 < x < 1\}$$
$$E_2 = (2,3) = \{x \text{ in } \mathbb{R} \text{ such that } 2 < x < 3\}$$
$$E_3 = (4,5) = \{x \text{ in } \mathbb{R} \text{ such that } 4 < x < 5\}$$

Notice the strict inequalities now (i.e. $<$ instead of $\leq$)! These points represent the edges of the triangle, To form our triangle T, we need to combine E_1, E_2, E_3 with the sets $\{1,2\}$, $\{3,4\}$ and $\{5,0\}$. We may just want to do:

$$T = \{E_1, E_2, E_3, \{1,2\}, \{3,4\}, \{5,0\}\}$$

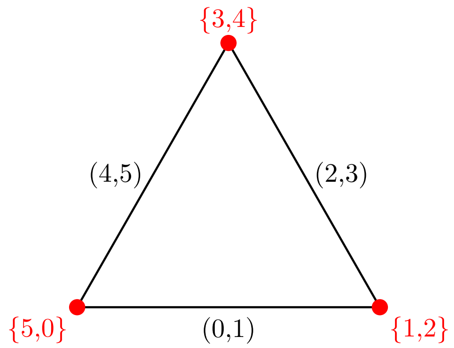
but E_1, E_2, E_3 are sets themselves, and hence T contains 6 elements representing 6 points. We shall need to be a little more careful with how we form T. To do this, we take the *union* of sets. The union of sets is represented by $\cup$ and is written as:

$$A \cup B = \{\text{take all } a \text{ in } A \text{ and all } b \text{ in } B\}$$

In other words, $A \cup B$ is a new set that containing all the elements of A and B. Define our triangle to be:

$$T = (0,1) \cup \{\{1,2\}\} \cup (2,3) \cup \{\{3,4\}\} \cup (4,5) \cup \{\{5,0\}\}$$

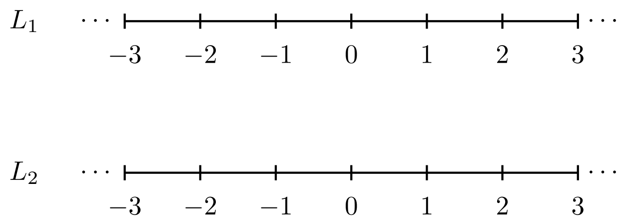
Let us break this down. T contains each element inside $(0,1)$, $(2,3)$ and $(4,5)$. It then also contains each element inside the three sets $\{\{1,2\}\}$, $\{\{3,4\}\}$, $\{\{5,0\}\}$. Each of these sets contains only a single element: they contain $\{1,2\}$, $\{3,4\}$, $\{5,0\}$. These are the glued vertices. We can visualize T as:



Giving us our Triangle!

Non-Hausdorff Space

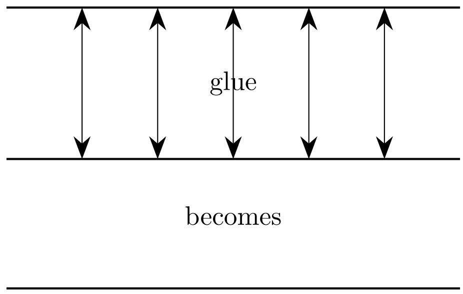
With this, we can finally construct our non-Hausdorff space. Let L_1 and L_2 be two copies of the real number line:



To not get confused with which numbers are we working with, let us label all numbers x in L_1 as x_1 and all numbers x in L_2 as x_2, for example 3_1 is in L_1 and 3_2 is in L_2. Let us now glue *all* the numbers x_1 in L_1 to the number x_2 in L_2 that has the same value, *except* for 0_1 and 0_2. So take all sets of numbers:

$$\{x_1, x_2\} \text{ (ex. } \{1_1, 1_2\}, \{32_1, 32_2\}, \text{ ...) but } 0_1, 0_2 \text{ are not in a set}$$

Taking the union of all these sets, we get a space N (N suggestively for "non-Hausdorff"). Let interpret N as a space. For any non-zero number, we glued them together:

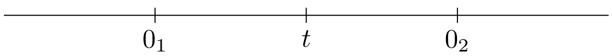


Thus, we don't need to distinguish between the nonzero points in N: we can just say 3 is in N as we glued $\{3_1, 3_2\}$. To be precise, we can label the set $\{3_1, 3_2\}$ as 3, and more generally if $x \neq 0$ we can say x is in N instead of saying $\{x_1, x_2\}$ is in N. As we have not glued the points 0_1 and 0_2, they remain distinct, so 0_1 and 0_2 are in N.

Now, we can finally ask:

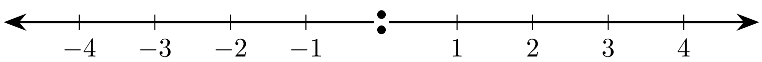
What is the distance between 0_1 and 0_2?

Let' say that there *was* some nonzero distance d between these points. Then we can find a nonzero number real number t that is between 0_1 and 0_2 and distance $d/2$ from either 0_1 or 0_2:



But that is absurd; there is *no* number between the zero's. Namely, the moment we try to move left from 0_1 or 0_2, we are encountering a negative number that is left from *both* numbers, and if we move right from 0_1 or 0_2, we are encountering a positive number that is right from *both* numbers, so there can't be a number between 0_1 or 0_2. This means the visual above is completely wrong; they have no distance separating them!

The space N is often attempted to be visualized as:



However this is misleading, as it looks like there is a gap between 0_1 and 0_2, but there is not. It is impossible to visualize this space as we live in a Hausdorff world[1]. Nevertheless, this image can be used to ground the intuition that there is no gap between 0_1 and 0_2; two points in the middle can be thought of as existing "one atop of another" in the same location.

[1] At least on the macro scale we live in

Any use?

Though this might seem like an exercise in arm-chair mathematising, these spaces do show up naturally in mathematics, especially within number theory and algebraic geometry. It also turns out that there are geometric properties of the world that do not need to consider spaces that are Hausdorff, leading to some interesting intuition about the nature of the idea of what it means to be "close". Unfortunately, to see why they show up *naturally* takes a lot of build up, and presenting them without this build-up will look a lot like random ideas stuck together. But this is also not too surprising as it took thousands of years before such spaces were found/invented, and then to realise that such spaces have immense practical properties in the 1950s (very recent mathematics!).

3 Technicalities

This section is to address the "white lies" presented above and is for those with some more technical background.

Defining a space as the set of numbers and subset of numbers is too vague; It doesn't define any notion of "locality" that is the key notion in topology, namely we have no notion of open or closed sets, and there are no restrictions on which sets can be part of a space (ex. we don't want the countable intersection of open sets in our collection, and for the euclidean topology we certainly don't want non-measurable sets to be open). Even worse, as every tuple is technically a set, the definition of space presented above includes $\mathbb{R}^2, \mathbb{R}^3, \ldots$, and in some model's of math can include any sets as all these space are ultimately sets of numbers, but it is "evident" that the definition is really only considering one-dimensional and 0-dimensional spaces. The first step to fixing this is by defining a topological space as a set of sets containing the empty set and itself, is closed under finite intersection, and arbitrary unions. We now have a notion of locality, and can start adding further constrictions such as being metrizable to actually have distance be well-defined. In fact, a metric cannot be defined on a non-Hausdorff space, and so the use of "distance" in the above is apocryphal. All details can be found in [1].

To not rely on distance, the Hausdorff condition is formally defined as:

Given two points $x, y \in X$ in a topological space X, there exists open sets U_x, U_y containing x, y respectively such that $U_x \cap U_y = \emptyset$

This shows that points can be *separated*. The approach taken here with the idea of distance is to avoid introducing the more abstract notion of "separation" and stick to the intuitive notion of distance.

The examples of glued sets almost respect the rules of an equivalence relationship; we technically should separate each number into a singleton set so that our space is a set of equivalence classes. The details can be found in [2].

References

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Topology*. 1st ed. 2022. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_topology.pdf.

[2] Nathanael Chwojko-Srawley. "Preliminary Chapter". In: (2023). URL: https://nathanaelsrawley.com/assets/pdfs/notes/preliminary_chapter.pdf.